Iterative Hardware Development of an Affordable Open-Source DPOAE Hearing Screening Device

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Motivation

Existing hearing screening devices cost **$7-10K**, putting them out of reach for many clinics. Our team is developing an **affordable, open-source hearing screening device** designed for use by **audiologists** and appropriately **trained community members**. The goal is to expand access to early hearing screening by supporting audiologists and diagnostic evaluation in areas where **access is limited**. The device is designed so that audiologists **can train community members** to perform preliminary hearing screenings and refer newborns with potential hearing loss back to them for diagnostic evaluation, **expanding access to early hearing screening**.

Design Goals

Affordable
Targeting 20–50% of the cost of existing devices for affordability.

Accessible
Designed for use by audiologists and appropriately trained community members.

Available
Built using off-the-shelf components and an open-source hardware architecture.

User Oriented
Developed with consideration for the needs of community-based hearing health systems.

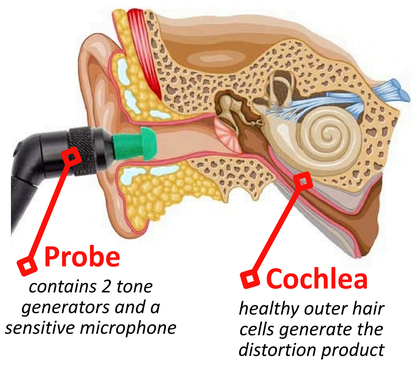
Giving Context

Why this Goal?
In many low-resource settings, newborn hearing screening is limited by expensive equipment and a lack of specialists. Children who miss early detection lose the window for timely language support.

Where our Insights are from?
Shaped by co-designs with the only audiologist in Guatemala, Dr Patricia Castellanos with a trained community member and partnership with Solar Ear.

Engagement with Stakeholders
We conduct user research through co-design sessions, where we listen to audiologists and community partners as co-designers. We then build their feedback into each iteration of the device and repeat the process.

DPOAE Screening



A diagnostic probe in the ear canal measuring cochlea responses
image credit: interacoustics.com

Distortion Product Otoacoustic Emissions (DPOAE) screening plays **two tones f_1 and f_2**, into the ear, causing healthy outer hair cells in the **cochlea** to generate a third tone called the distortion product, **$f_{DP} = 2f_1 - f_2$**. A sensitive probe microphone **measures this response** and compares it against the surrounding noise floor. A detectable response indicates **healthy cochlear function** and produces a **PASS**, while a **weak or absent response** produces a **REFER** for further evaluation.

Methods

Detecting a DPOAE requires capturing a **very small acoustic response** with sufficient signal quality. We are **evaluating analog and digital microphones** to determine which best captures this response. The Rev. 3 board **integrates both pathways** on one platform for **comparison and debugging.**

Design for Testability on the Rev. 3 Board

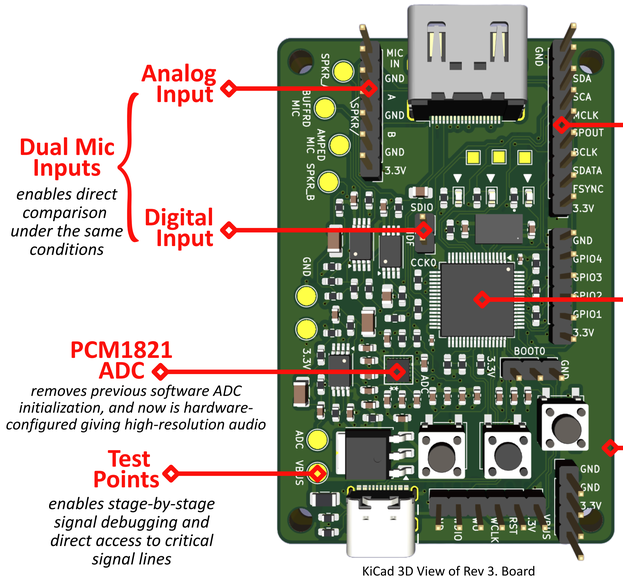
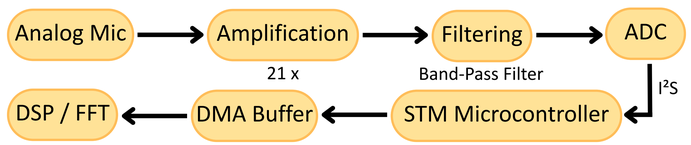


image credit: DJS Global Health

KiCad 3D View of Rev 3. Board

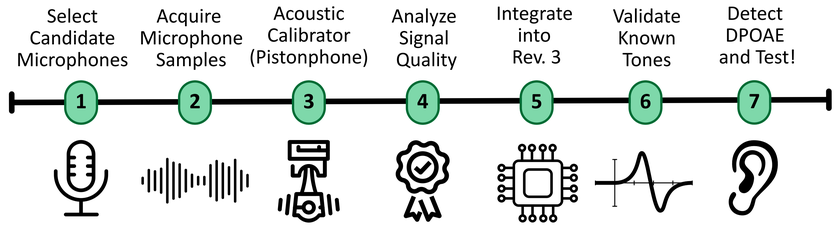
Analog Microphone Pipeline



Digital Microphone Pipeline



Development & Testing Workflow



Results & Conclusion



image credit: Ximena Muñoz, Fundación Sonrisas que Escuchan

Technician performing a newborn hearing screening in a hospital in Guatemala

Previous hardware iterations showed that reliable DPOAE acquisition requires both sufficient microphone **signal quality** and a systematically **testable architecture**. The **Rev. 3 platform applies these lessons** by enabling independent evaluation of analog and digital microphone pathways to determine **which approach can meet DPOAE requirements**.
These technical decisions also support the broader goal of an **affordable, open-source** device. Using **off-the-shelf components and testable hardware** enables partners to **fabricate, troubleshoot, manufacture, and adapt** the device for their specific contexts.

Next Steps

Steps	Details & Specifics
Board Bring-Up	1 — Verify power, clocks, PCM1821/I²S communication, MDF/PDM acquisition, and DMA data transfer for both microphone pathways.
Microphone Characterization	2 — Integrate selected analog and digital microphones to quantify frequency response, noise, and SNR across DPOAE-relevant frequencies.
Compare Acquisition Pipelines	3 — Evaluate analog and digital pathways under consistent test conditions to determine which provides the signal quality for DPOAE detection.
End-to-End DPOAE Testing	4 — Generate the two-tone f_1 and f_2 stimulus, capture the acoustic response in a test coupler, and evaluate detection of $(2f_1 - f_2)$ distortion product.
Refine the Device	5 — Use validation results to select the pipeline and acquisition architecture for future revisions and continue integration into the handheld device.

Acknowledgements

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